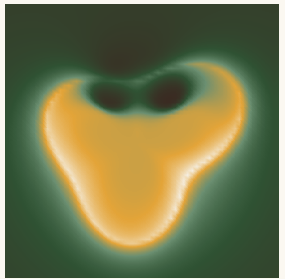
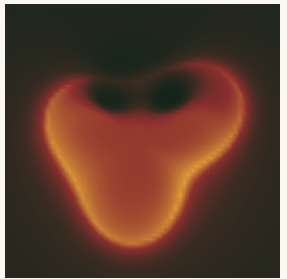
# One field, five readings

*Totoro*



*Shishigami*



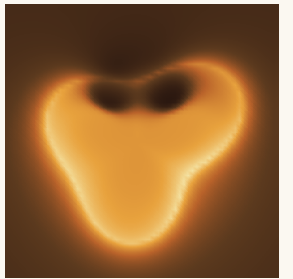
*Kodama*



*No-Face*

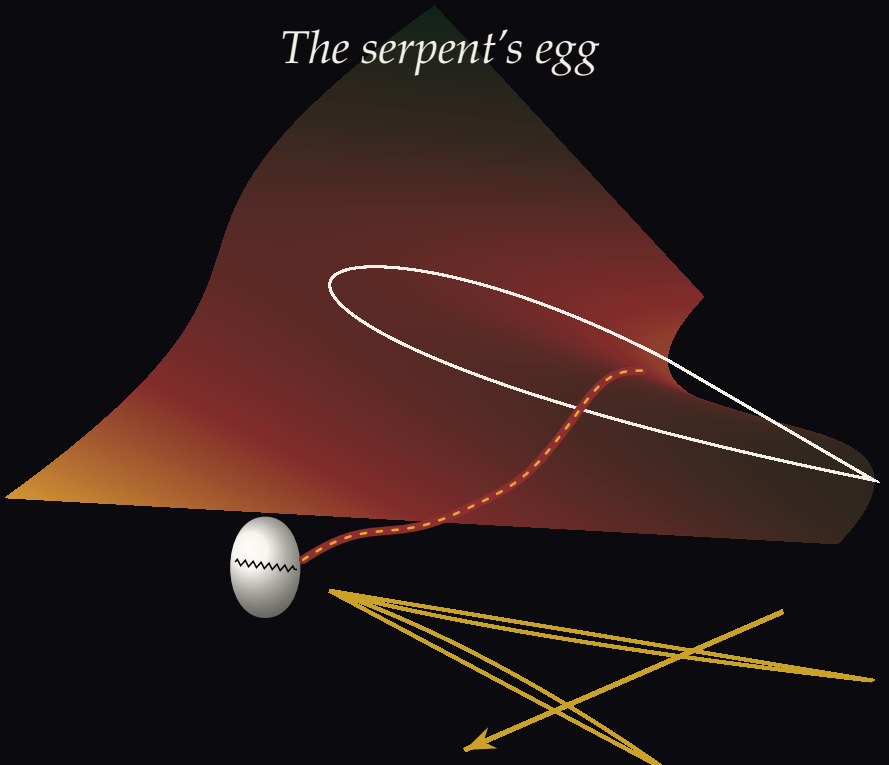


*Catbus*



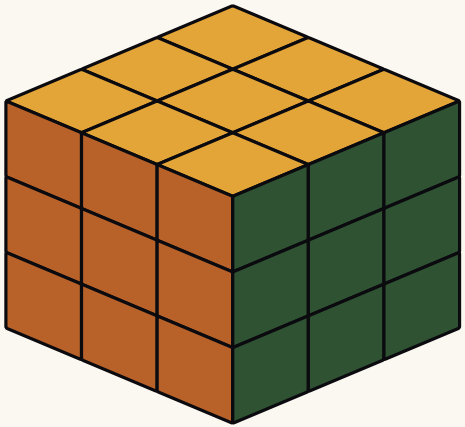
$\Phi = e^{-\lambda|\rho - r(\theta)|} + c_I\big(1 + e^{\beta(\rho - r(\theta))}\big)^{-1} + c_M \tanh \sum_i \alpha_i / (|x - x_i|^2 + \varepsilon)$, $r(\theta) = 1 + 0.22\sin 3\theta + 0.12\sin(2\theta + 0.35)$: the same scalar field under the five palettes.

# *The serpent's egg*

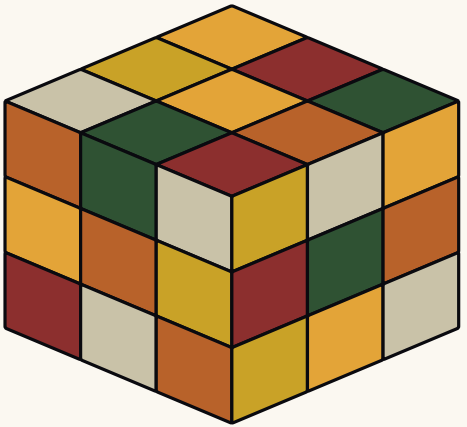


The sheet $x^3 + ux + v = 0$ over the control plane $(u, v)$, folded along $3x^2 + u = 0$; its shadow on the floor is the cusp $4u^3 + 27v^2 = 0$. A path that crosses the cusp jumps from one sheet to the other: the latent form hatches.

# *A cube in five palettes*



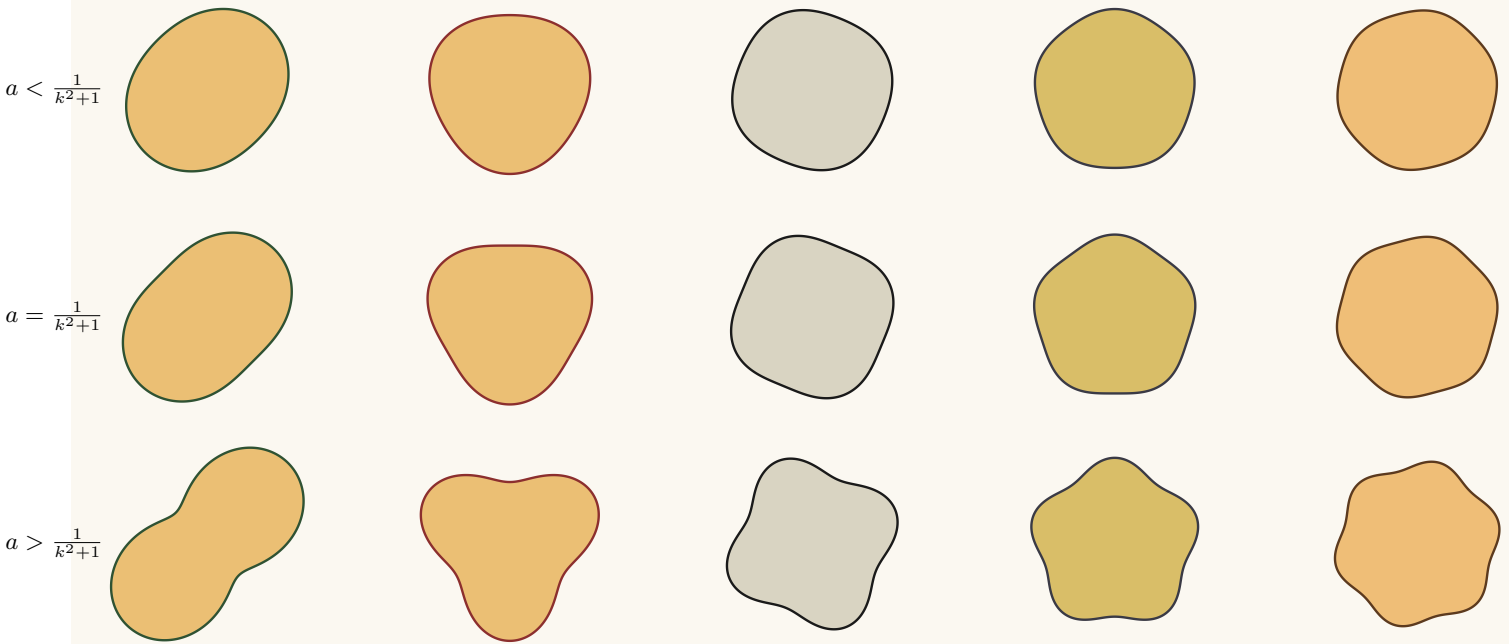
*solved: one colour per face*



*after twists: the colours wander, the invariants do not*

Six face colours from the Totoro, Catbus, Shishigami, Kodama and No-Face palettes. Every legal position keeps the edge-permutation parity equal to the corner-permutation parity, the corner twists summing to $0 \bmod 3$ and the edge flips to $0 \bmod 2$: one reassembly in twelve is solvable.

# *The threshold of convexity*




$r(\theta) = 1 + a \sin k\theta$ is convex exactly when $a \leq 1/(k^2+1)$: the middle row has $k$ flat points, the bottom row $2k$ inflections.